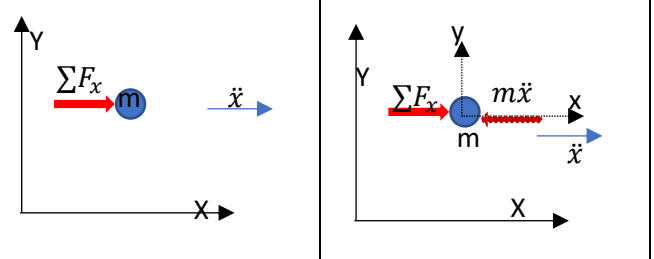


D'ALEMBERT'S PRINCIPLE

Consider a particle of mass m moving with an acceleration $\ddot{x}$. An observer sitting on the reference X-Y axes will see the particle moving with acceleration $\ddot{x}$. Then we can write the equation of motion as $\sum F_x = m\ddot{x}$.



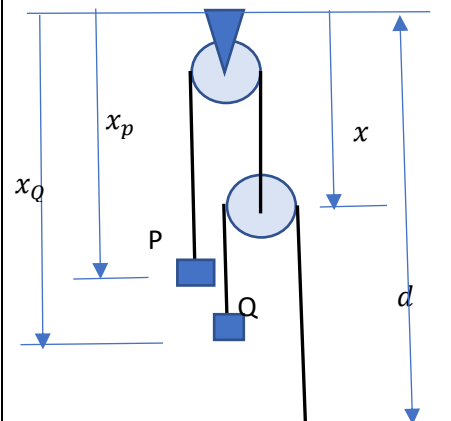
Assume that another observer is sitting on the co-ordinate x-y axes which is moving along with the particle. To that observer, the particle will appear to be in equilibrium though he can see the force $\sum F_x$ is acting. This situation can be visualized as a **dynamic equilibrium**. This arises when a force $m\ddot{x}$ is assumed to act in the opposite direction of acceleration. Then for the observer on the body $\sum F_x - m\ddot{x} = 0$. This statement is known as **D'Alembert's principle**. By this principle, a force $m\ddot{x}$ is applied to the body in a direction opposite to the direction of acceleration (known as **inertia force**) and we can write that the algebraic sum of all forces is equal to zero. We can reduce a dynamic problem into an equivalent statics problem. Please remember that this is only a technique by which we can solve problems and is another way of writing the Newton's law.

Since, when a body is translating all the infinitesimally small mass points of that body will have the same acceleration, the inertia forces of all these mass points will constitute a system of parallel distributed forces. Hence resultant inertia force will act through the centre of mass.

Example 1

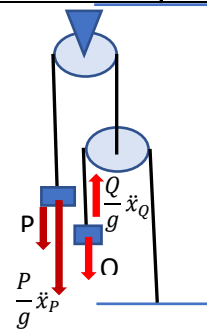
Neglecting friction and inertia of the two pulleys, find the acceleration of the falling weight Q assuming $P=Q$

Let the weight Q is moving down with an acceleration $\ddot{x}_Q$. The weight P will move up by an acceleration $\ddot{x}_P$. Then, by D'Alembert's principle inertia forces of magnitude $\frac{Q}{g}\ddot{x}_Q$ and $\frac{P}{g}\ddot{x}_P$ will act in the directions opposite to the respective acceleration. Then the system can be considered as in a dynamic equilibrium. Then, we can apply the principle of virtual work to solve this problem.



In principle of virtual work, we will not consider the internal forces. Hence tension in the strings are not considered. Let us give a small virtual displacement dx_Q to the weight Q downwards. Then P will move dx_P up. Then by the Principle of Virtual work, algebraic sum of the virtual work done by forces is equal to zero when the system is in "equilibrium".

$$Q \cdot dx_Q - \frac{Q}{g} \ddot{x}_Q \cdot dx_Q - P \cdot dx_P - \frac{P}{g} \ddot{x}_P \cdot dx_P = 0$$



Now we have to develop a relation connecting the accelerations of P and Q. Assuming each string to be inextensible, we can consider the length of the string to be constant.

Considering the string carrying weight P, we can write, $x_P + x = \text{constant}$

Considering the string carrying weight Q, $x_Q - x + d - x = \text{constant}$, or $x_Q - 2x = \text{constant}$

Differentiating the two expressions, $dx_P + dx = 0 \gg \gg dx_P = -dx$

$$dx_Q - 2dx = 0 \gg \gg dx_Q = 2dx = -2dx_P$$

The negative sign shows that the differential change in the positions of P and Q are opposite.

If we differentiate the above expression with respect to time

$$\frac{dx_q}{dt} = -2 \frac{dx_p}{dt}$$

Differentiating once again with respect to time

$$\frac{d^2x_q}{dt^2} = -2 \frac{d^2x_p}{dt^2}$$

The acceleration of Q is twice the acceleration of P in opposite direction. The directions are already taken care of.

$$\ddot{x}_Q = 2\ddot{x}_P$$

Now, the equation of virtual work becomes

$$Q \cdot dx_Q - \frac{Q}{g} \ddot{x}_Q \cdot dx_Q - P \cdot \frac{dx_Q}{2} - \frac{P}{g} \cdot \frac{\ddot{x}_Q}{2} \cdot \frac{dx_Q}{2} = 0$$

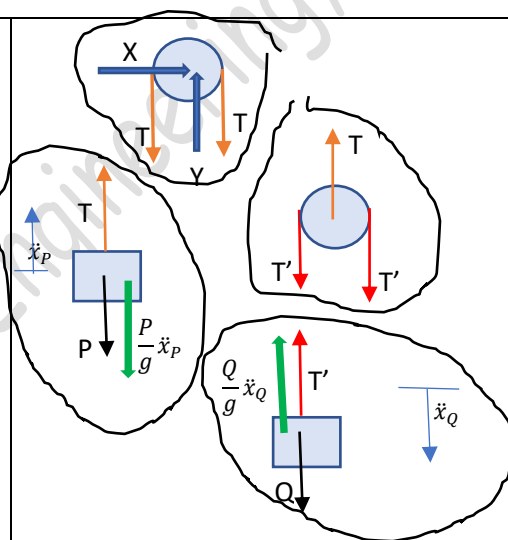
$$\ddot{x}_Q = \frac{2}{5}g$$

We can also solve this problem without virtual work principle. Now we draw the free body diagram of the system. Inertia forces of $\frac{P}{g}\ddot{x}_P$ and $\frac{Q}{g}\ddot{x}_Q$ are incorporated in the free-body diagrams of blocks P and Q in directions opposite to the respective accelerations. Since the inertia of the pulleys are neglected, the moving pulley will not have any acceleration and hence no inertia force. Applying D'Alembert's principle to block P

$$T - P - \frac{P}{g}\ddot{x}_P = 0$$

Applying D'Alembert's principle to block Q

$$\frac{Q}{g}\ddot{x}_Q + T' - Q = 0$$



For the moving pulley

$$T - 2T' = 0$$

Solving these equations along with $\ddot{x}_Q = 2\ddot{x}_P$, we get the values of $\ddot{x}_Q, \ddot{x}_P, T$, and T'

$$2T' - P - \frac{P}{g} \frac{\ddot{x}_Q}{2} = 0$$

$$2 \frac{Q}{g} \ddot{x}_Q + 2T' - 2Q = 0$$

$$\frac{\ddot{x}_Q}{g} \left(2Q + \frac{P}{2} \right) = (2Q - P)$$

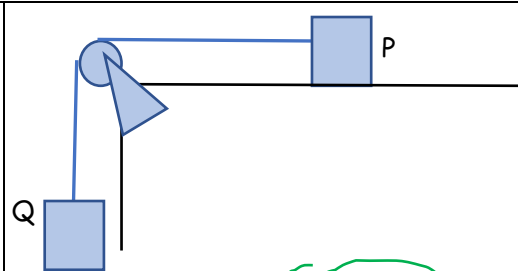
$$\ddot{x}_Q = \frac{2}{5}g$$

$$T' = Q \left(1 - \frac{2}{5} \right) = \frac{3Q}{2} \gggg T = 3Q$$

We can also solve this problem by Newton's equation. Also note the values of tensions in the strings. Compare this values with those when the system is in static equilibrium.

Example 2

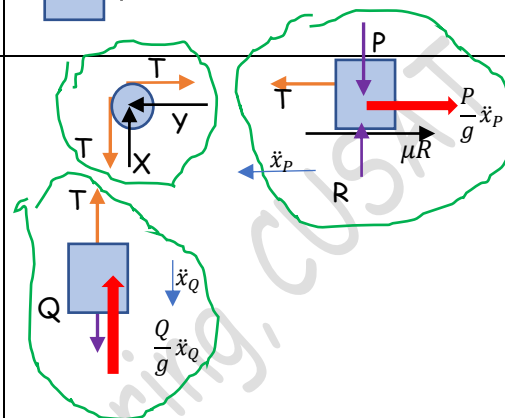
Two blocks of weights $P=54\text{ N}$ and $Q=27\text{ N}$ are connected by a flexible but inextensible cord and supported as shown. If the coefficient of friction between the block P and surface is 0.3 . Neglecting friction and inertia of the pulley, find the acceleration of the system and the tension in the string



Since the string is inextensible, the displacements of blocks P and Q are equal in magnitude. Hence, the velocity and acceleration of both blocks are the same in magnitude.

$$\ddot{x}_Q = \ddot{x}_P$$

Incorporating the inertia forces in the direction opposite to the direction of acceleration, the system is considered to be in dynamic equilibrium.



From block P :

$$R - P = 0$$

$$\frac{P}{g} \ddot{x}_P + \mu R - T = 0 \quad \gg \gg \quad \frac{P}{g} \ddot{x}_P + \mu P - T = 0$$

From block Q :

$$T + \frac{Q}{g} \ddot{x}_Q - Q = 0$$

From these equations we can write

$$\frac{P}{g} \ddot{x}_P + \mu P - T + T + \frac{Q}{g} \ddot{x}_Q - Q = 0$$

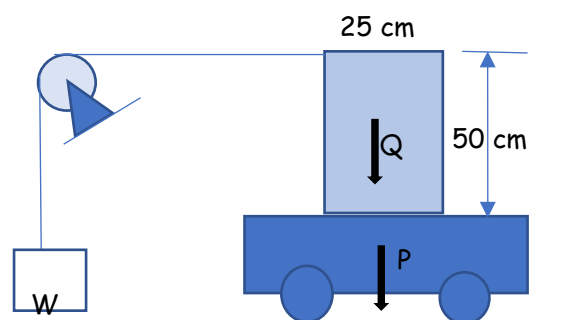
$$(P + Q) \frac{\ddot{x}}{g} = Q - \mu P$$

$$\ddot{x} = \frac{27 - 0.3 \times 54}{54 + 27} g = 0.133 g$$

$$T = 27 - \frac{27}{g} \times 0.133 g = 23.4 \text{ N}$$

Example 3

A rectangular block of weight $Q=890\text{ N}$ rests on a flat car of weight $P=445\text{ N}$ which may roll along a horizontal plane without friction. The car and the block together are to be accelerated by a weight arranged as shown. Assuming that there is sufficient friction between the block and the car to prevent sliding, find the maximum weight W by which the car can be accelerated. What will be the magnitude of this maximum acceleration? Neglect the friction and inertia of pulley.



Let W be the maximum weight which will give a maximum acceleration of $\ddot{x}$ to the car and block. Since the friction between the block and the car is sufficient to prevent sliding, when the acceleration is maximum the block will tend to tip over the left edge. Draw the free body diagrams of the system incorporating the inertia forces

From the weight W :

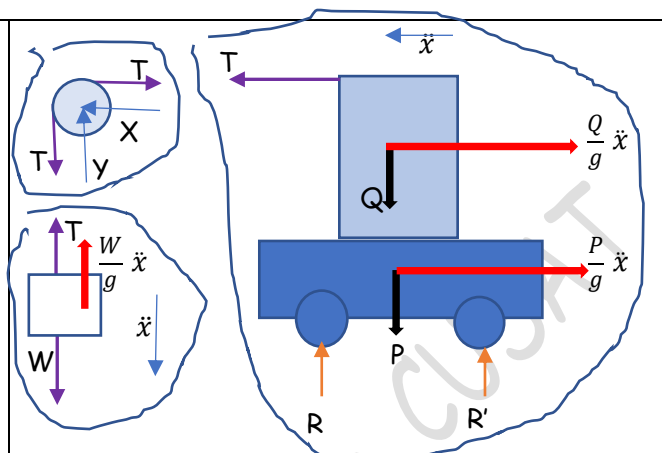
$$\frac{W}{g} \ddot{x} + T - W = 0 \dots \dots (1)$$

From the car and block:

$$\frac{Q}{g} \ddot{x} + \frac{P}{g} \ddot{x} - T = 0 \dots \dots (2)$$

$$R + R' - P - Q = 0$$

We have three unknowns (W , $\ddot{x}$, T) in the first two equations and third one does not have any relevance to this problem. Let us try to find out one more equation by considering the free body diagram of the block Q



Writing the equation $\Sigma M_A = 0$

$$T \times 50 \text{ cm} - Q \times 12.5 \text{ cm} - \frac{Q}{g} \ddot{x} \times 25 \text{ cm} = 0 \dots \dots (3)$$

$$(1) \gg \gg \gg T = W \left(1 - \frac{\ddot{x}}{g} \right)$$

$$(2) \gg \gg \gg W \left(1 - \frac{\ddot{x}}{g} \right) = (P + Q) \frac{\ddot{x}}{g} \gg \gg \ddot{x} = \frac{W}{W + P + Q} g = \frac{W}{W + 1335} g$$

$$(3) \gg \gg W \left(1 - \frac{\ddot{x}}{g} \right) \times 50 = Q \times 12.5 + \frac{Q}{g} \ddot{x} \times 25$$

$$(50W + 25Q) \frac{\ddot{x}}{g} = 50W - 12.5Q$$

$$\ddot{x} = \frac{50W - 12.5Q}{50W + 25Q} g$$

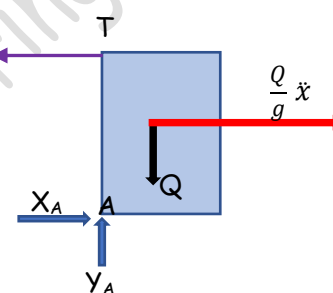
Therefore, we can write

$$\frac{W}{W + 1335} g = \frac{50W - 12.5Q}{50W + 25Q} g$$

$$50W^2 + 22250W = 50W^2 + 66750W - 11125W - 14851875$$

$$W = 445 \text{ N}$$

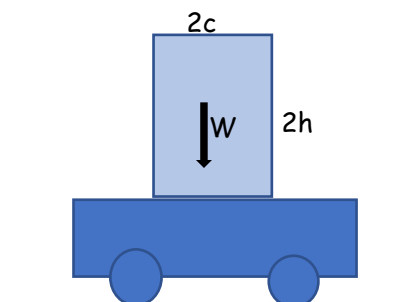
$$\ddot{x} = 0.25g$$



Exercise 1

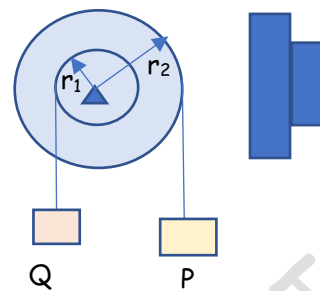
A block of weight W , height $2h$, and width $2c$ rests on a flat car which moves horizontally with constant acceleration a . Determine the value of acceleration a at which slipping of the block on the car will impend if the coefficient of friction is μ .

If there is sufficient friction to prevent slipping, at what acceleration tipping of the block will impend?



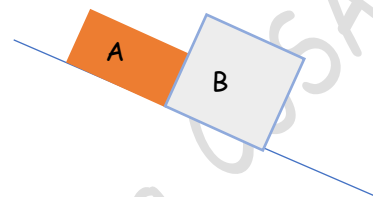
Exercise 2

Neglecting friction and inertia of the step-pulley, find the acceleration of the falling weight P . Given $P = 400\text{ N}$, $Q = 600\text{ N}$ and $r_2 = 2r_1$.



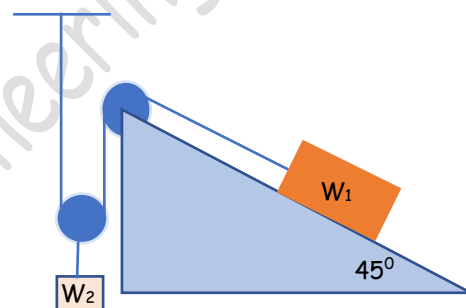
Exercise 3

Two blocks of weights $W_A = 500\text{ N}$ and $W_B = 1000\text{ N}$ slide down a 30° inclined plane under the action of gravity. If the coefficient of friction between them and the inclined plane are $\mu_A = 0.15$ and $\mu_B = 0.30$, find the force existing between the blocks during motion.



Exercise 4

Find the tension in the string during motion of the system shown if $W_1 = 1000\text{ N}$ and $W_2 = 500\text{ N}$. The coefficient of friction between the inclined plane and the block is 0.2. Neglect the friction and inertia of the pulleys.



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